Kernix 0.13.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

51 workloads

175 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.13.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ

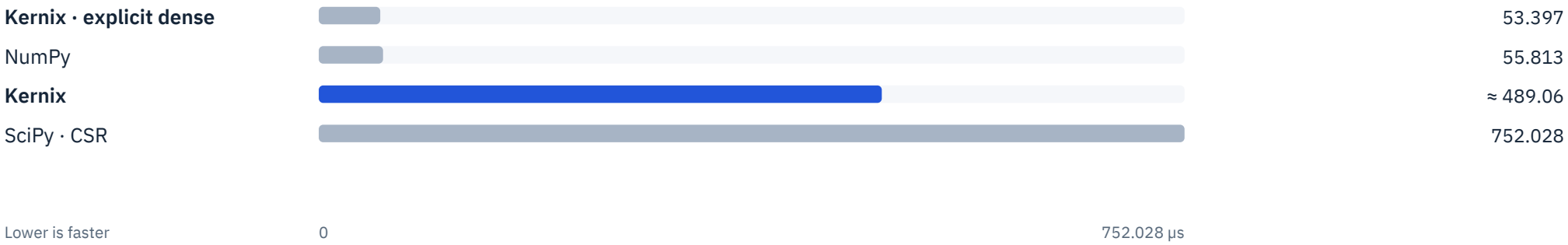


Kernix 0.13.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



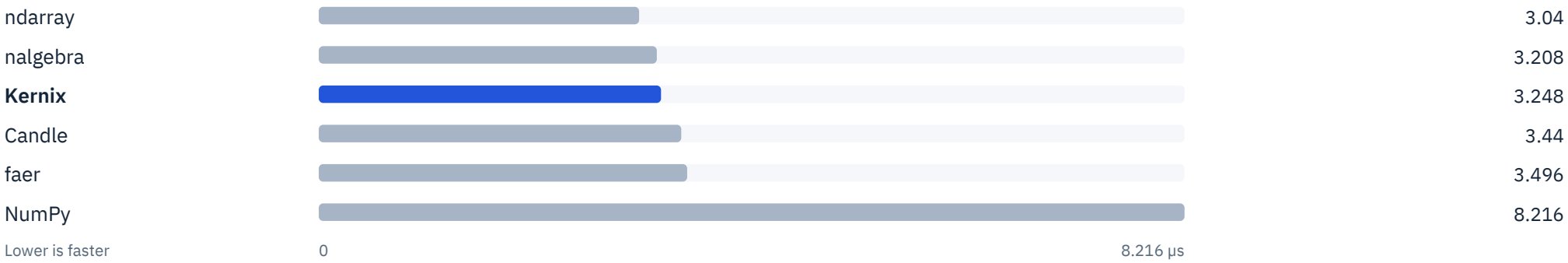
Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

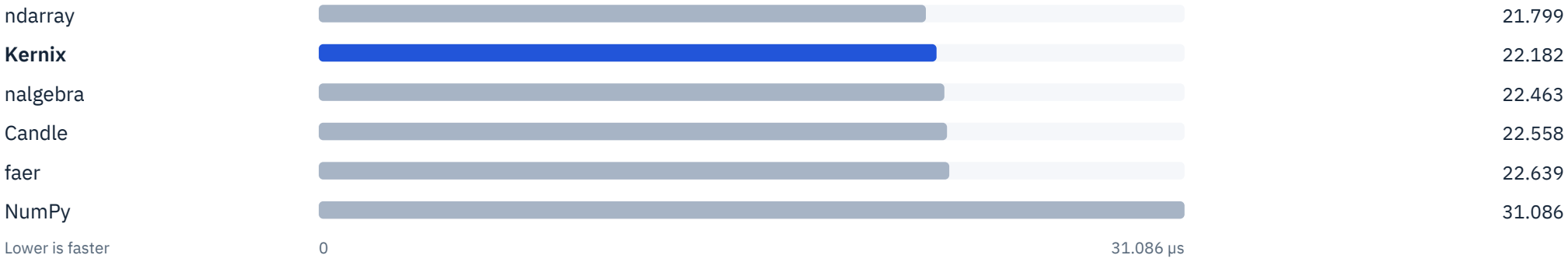
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



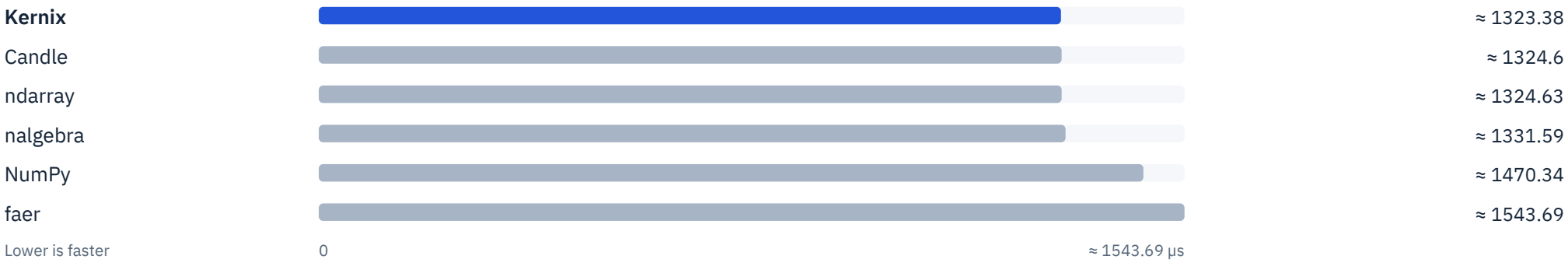
Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

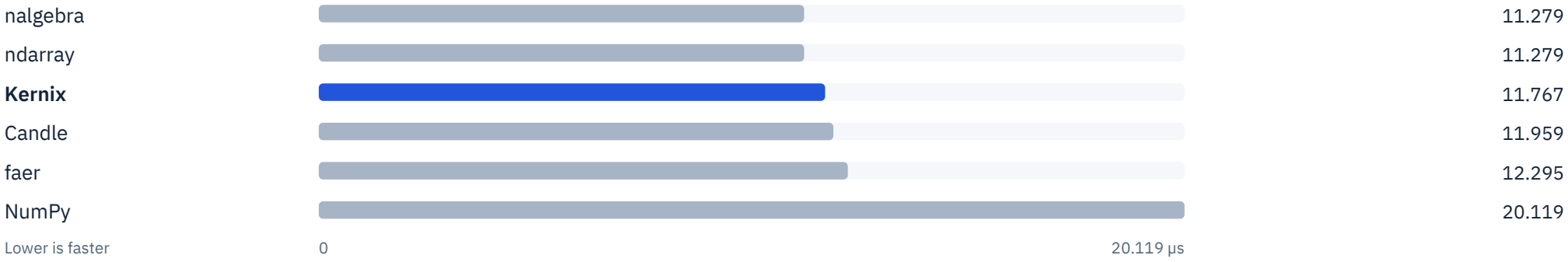
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



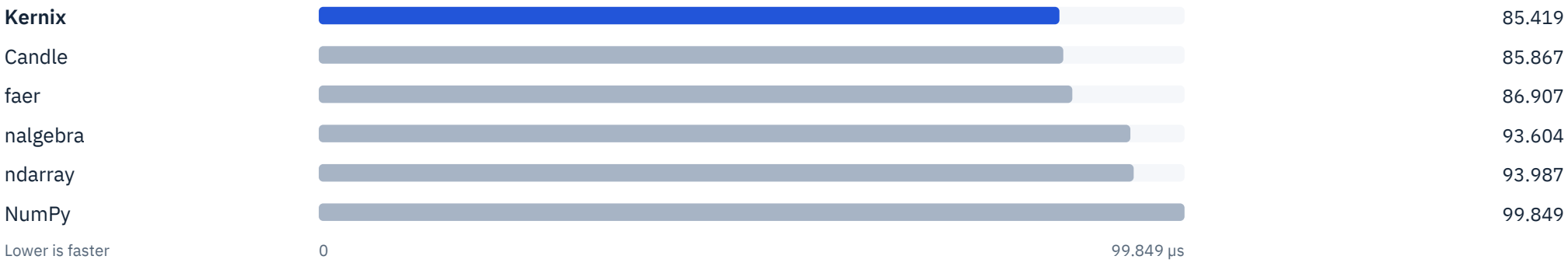
Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

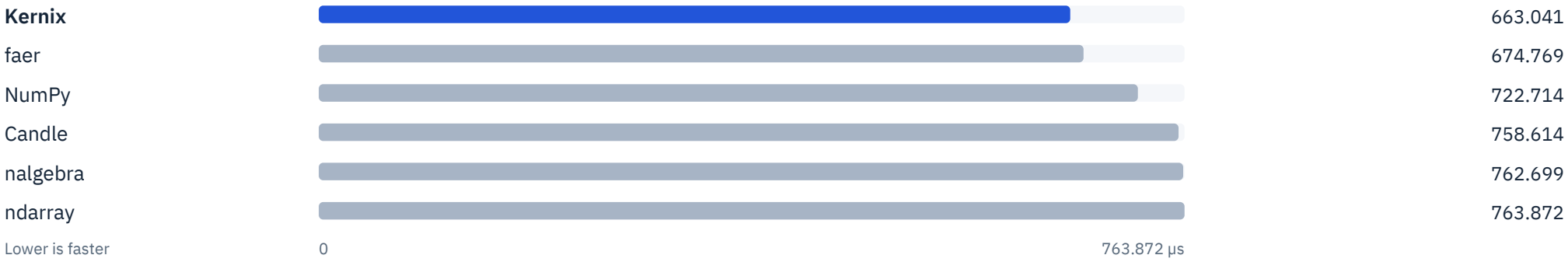
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.13.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

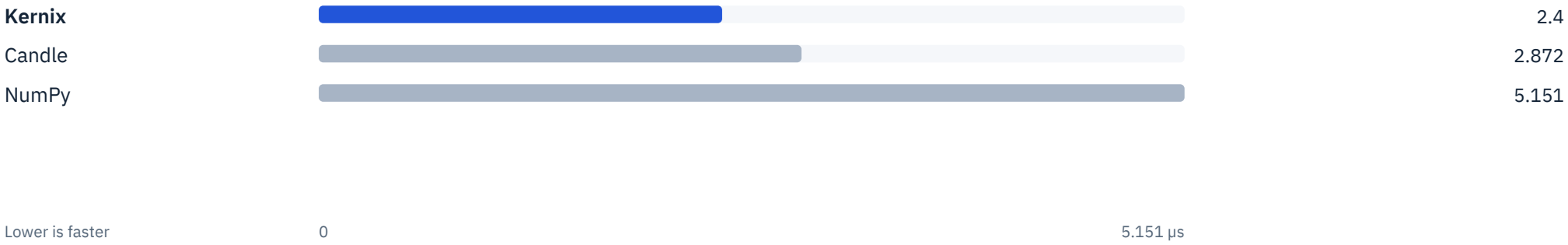
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.13.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



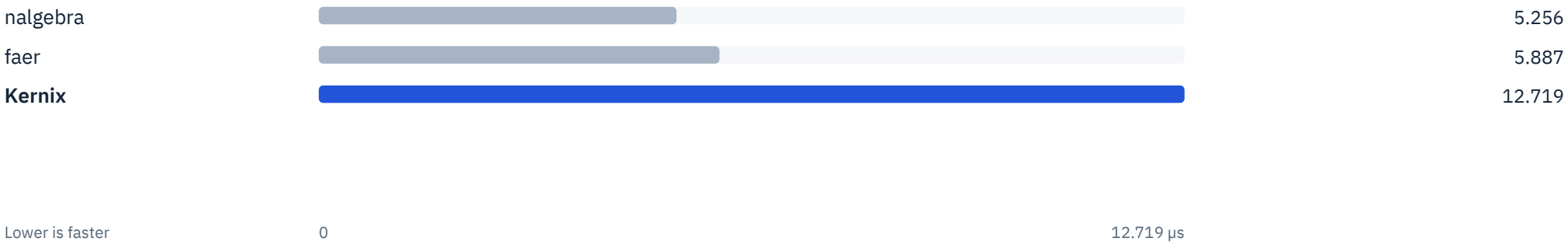
Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

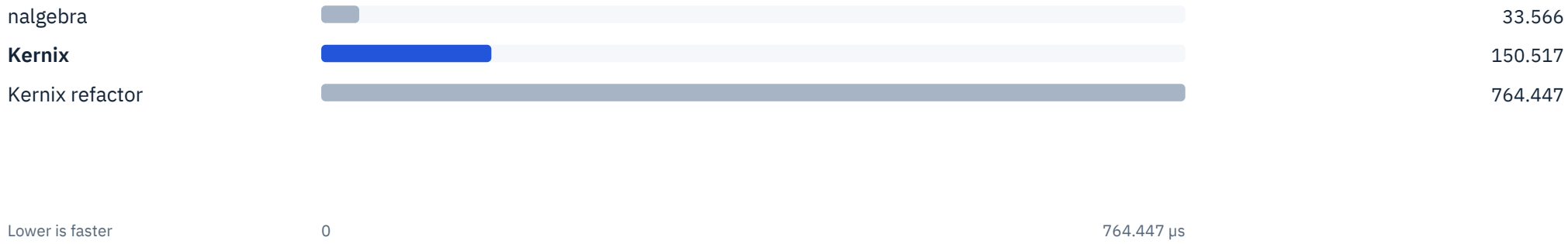
Cholesky rank-one update

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

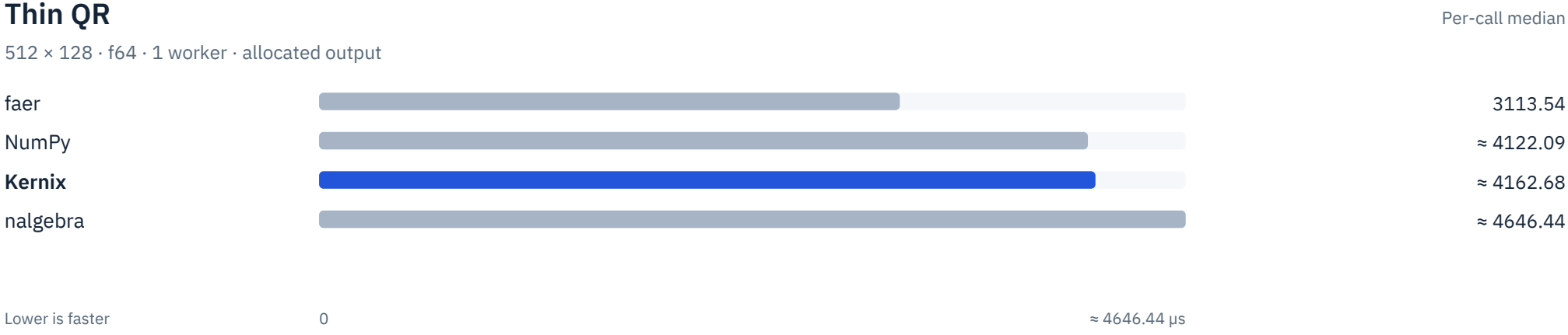
Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output



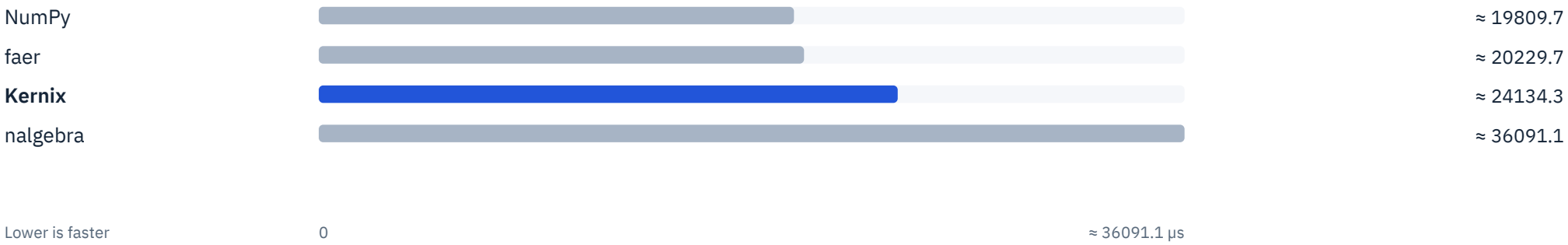
Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

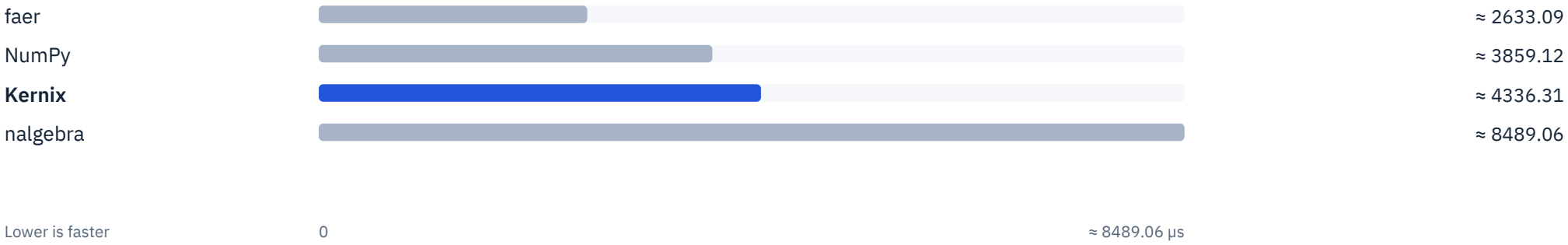
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



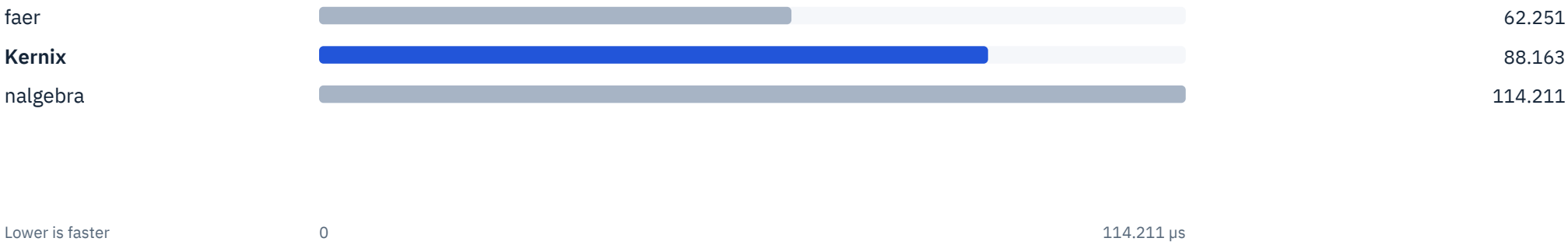
Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

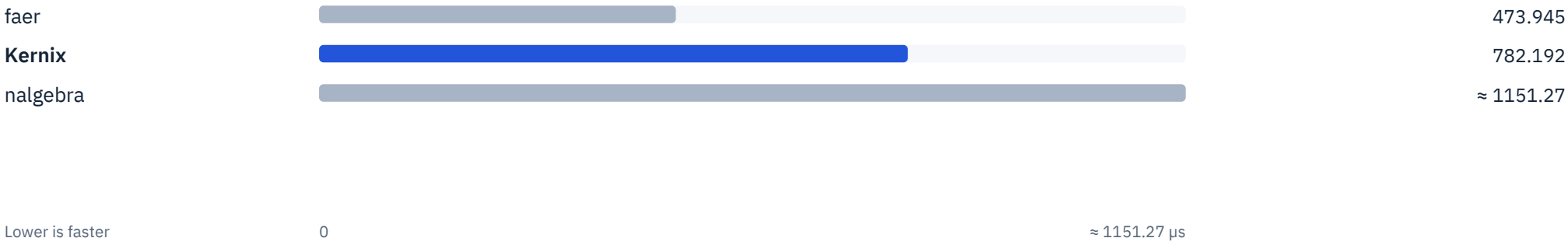
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



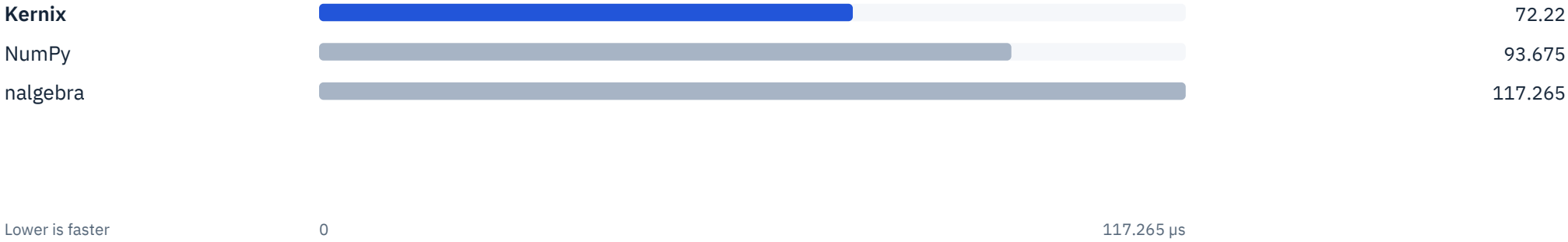
Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

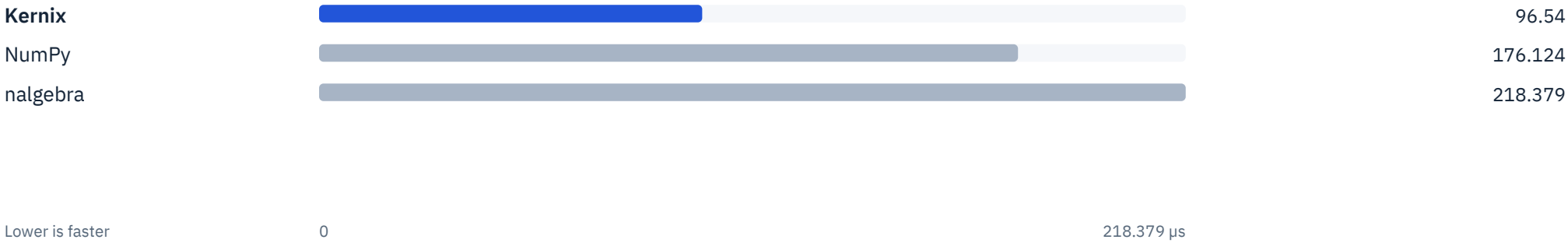
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

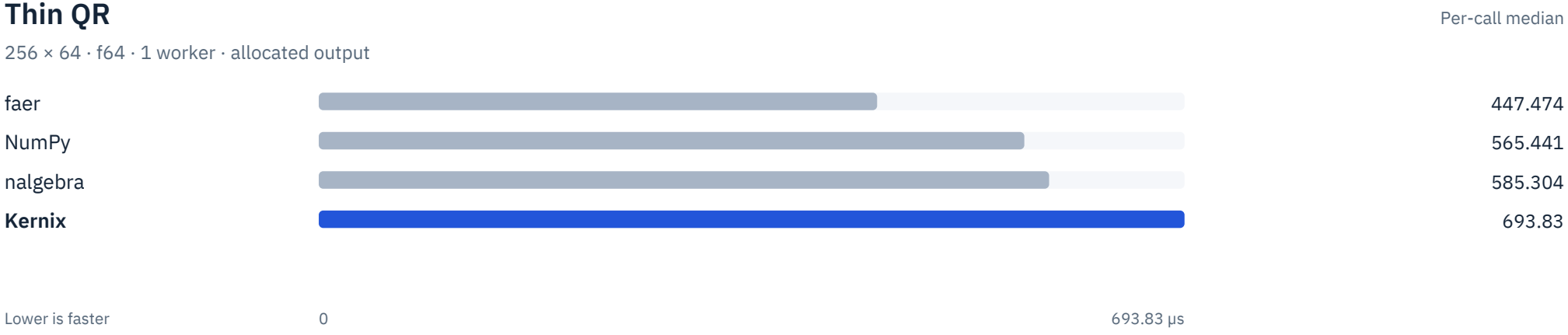
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.13.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



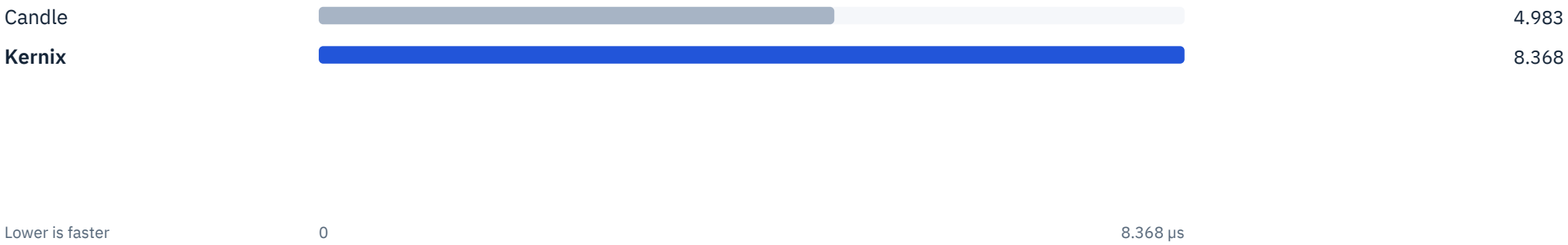
Kernix 0.13.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.13.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

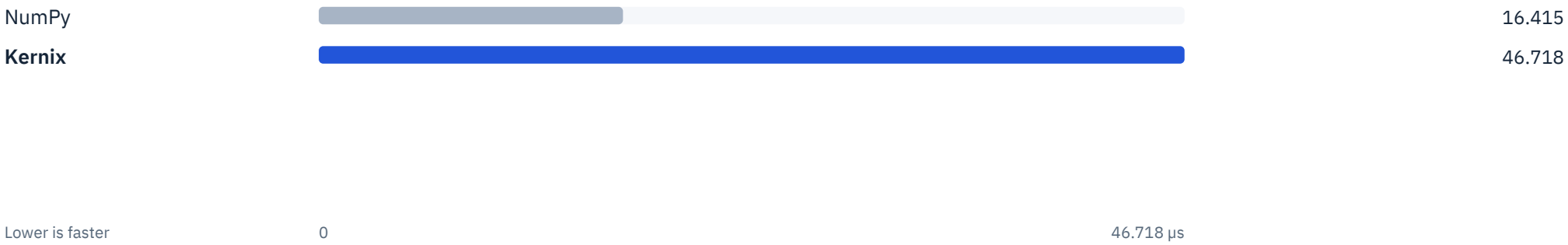


Kernix 0.13.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

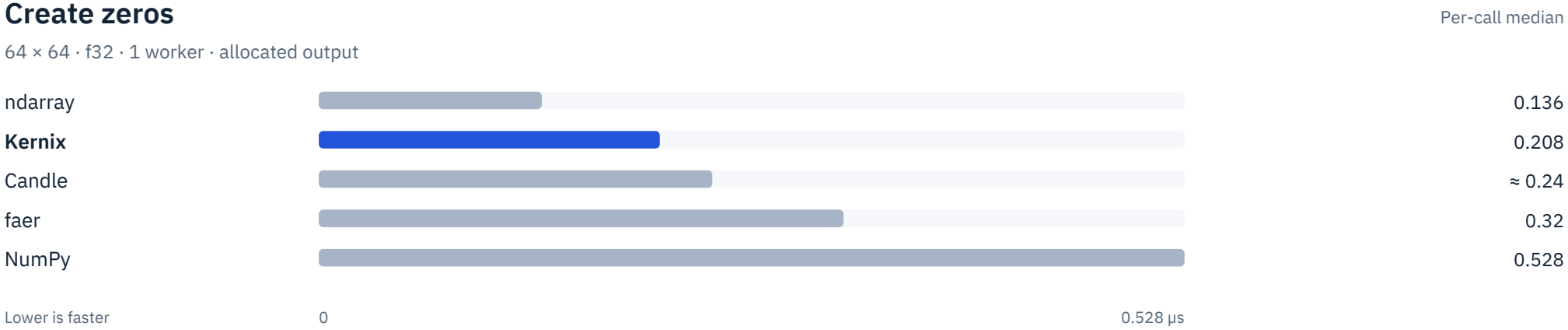


Kernix 0.13.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0209528362014022	1.0044167521097878 – 1.028552992418132

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9999389678238367	0.9988822449399083 – 1.0019213770644335

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.999545442181394	0.9989342808541711 – 1.0014338340055817

Thin QR

512 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0013490003755645	0.9992072846344657 – 1.0035722503949913

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9999680183567204	0.998100644669221 – 1.006039028938504

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9932030905284503	0.9862801868789893 – 1.007483188743173

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9959944736588449	0.9920063778899814 – 0.9988831974914898

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9947204933445353	0.9924569352137417 – 1.0028312037757108

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0003095017022594	0.9996913580246913 – 1.0024752475247525

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.002187844645146	0.9990965717274499 – 1.0042834705630383

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9988776130571014	0.9974802855676357 – 1.0012975171833356

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0000733170524125	0.9988657096941796 – 1.0012095566320618

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0003351722513611	0.9982475085060689 – 1.0032564327396971

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0022779043280183	1.0002841716396704 – 1.0113895216400912

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0021819561711414	0.992643247573273 – 1.0048934706273134

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9977473271394556	0.9967462797966133 – 1.0013204173921173

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0020833333333337	0.9979209979209976 – 1.004428236519927

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0033444816053512	1.000418235048097 – 1.0033444816053512

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0025886337406835	0.9989399875376913 – 1.0068572085058107

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9943369422082682	0.9896212753455892 – 1.001343488214958

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9991390368849621	0.994341539844011 – 1.0006276015852136

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9968366943455912	0.9819406051349345 – 1.0067972337839728

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9989854787615325	0.9964796177870741 – 1.008395017412424

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9975581228325775	0.9960705660840735 – 1.000474285478616

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	8016.000000000001	8.016000000000001
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	9760	9.76
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	805888	805.888
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2914433	2914.433
Sparse matrix × vector / Kernix 2048 × 2048 · five diagonals · f64	18142	18.142
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	23047	23.047
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	783657	783.657
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	1064764	1064.764
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	68052	68.052
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f64	30454	30.454

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	15343	15.343
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f32	21079	21.079
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	380793	380.793
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	512470	512.47
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	64019	64.019
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	1475794	1475.794
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	47037	47.037
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	1324545	1324.545
Sparse × sparse matrix / Kernix 256 × 256 · five diagonals each · f64	64338.99999999999	64.33899999999999
Automatic CSR construction / Kernix 512 × 512 · clone + scan + compress · f64	489060.00000000006	489.06000000000006

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	752028	752.028
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	53397	53.397
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	55813	55.813
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	195461	195.461
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	214161	214.161
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	93163	93.163
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	100170	100.17
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	168693	168.693
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	195731	195.731
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	100042	100.042

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	85468	85.468
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3248	3.248
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3496	3.496
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8216	8.216
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3040	3.04
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3440	3.44
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3208	3.208
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21799	21.799
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22463	22.463
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22182	22.182

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22639	22.639
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22558	22.558
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31086	31.086
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	209087	209.087
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	196046	196.046
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169638	169.638
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	168406	168.406
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	209276	209.276
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168223	168.223
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1323383	1323.383

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1324634	1324.634
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1470338	1470.338
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1543694	1543.694
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1324601	1324.601
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1331586	1331.586
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20119	20.119
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11767	11.767
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12295	12.295
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11959	11.959
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11279	11.279

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11279	11.279
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93604	93.604
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93987	93.987
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85867	85.867
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85419	85.419
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	86907	86.907
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99849	99.849
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	763872	763.872
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	674769	674.769
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	762699	762.699

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	758614	758.614
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	722714	722.714
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	663041	663.041
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4408	4.408
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12999	12.999
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3831.0000000000005	3.8310000000000005
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5151	5.151
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2872	2.872
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2400	2.4
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16448	16.448

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	93930	93.93
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	692783	692.783
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	50333	50.333
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	12719	12.719
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5887	5.887
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5256	5.256
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	452655	452.655
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1719759	1719.759
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1286693	1286.693
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	3193317	3193.317

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	12850165	12850.165
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	13394223	13394.223
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	457732	457.732
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1654199	1654.199
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1234129	1234.129
Cholesky rank-one update / Kernix 256 × 256 · f64	151119	151.119
Cholesky rank-one update / nalgebra 256 × 256 · f64	33846	33.846
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	764425	764.425
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	33566	33.566
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	764447	764.447

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one downdate / Kernix 256 × 256 · f64	150517	150.517
Cholesky rank-one update / nalgebra 512 × 512 · f64	148319	148.319
Cholesky rank-one update / Kernix 512 × 512 · f64	657513	657.513
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4356637	4356.637
Thin QR / NumPy 512 × 128 · f64	4122093	4122.093
Thin QR / Kernix 512 × 128 · f64	4162684	4162.684
Thin QR / nalgebra 512 × 128 · f64	4646439	4646.439
Thin QR / faer 512 × 128 · f64	3113540	3113.54
Thin QR / nalgebra 1024 × 256 · f64	36091080	36091.08
Thin QR / faer 1024 × 256 · f64	20229733	20229.733

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 1024 × 256 · f64	24134286	24134.286
Thin QR / NumPy 1024 × 256 · f64	19809741	19809.741
Cholesky factorization / Kernix 512 × 512 · f64	4336307	4336.307
Cholesky factorization / NumPy 512 × 512 · f64	3859118	3859.118
Cholesky factorization / nalgebra 512 × 512 · f64	8489056	8489.056
Cholesky factorization / faer 512 × 512 · f64	2633094	2633.094
Column-pivoted QR / nalgebra 96 × 32 · f64	114211	114.211
Column-pivoted QR / Kernix 96 × 32 · f64	88163	88.163
Column-pivoted QR / faer 96 × 32 · f64	62251	62.251
Column-pivoted QR / Kernix 256 × 64 · f64	782192	782.192

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 256 × 64 · f64	1151266	1151.266
Column-pivoted QR / faer 256 × 64 · f64	473945	473.945
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	117265	117.265
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72220	72.22
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93675	93.675
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96540	96.54
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218379	218.379
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176124	176.124
Thin QR / Kernix 96 × 32 · f64	73244	73.244
Thin QR / nalgebra 96 × 32 · f64	61291.99999999999	61.29199999999999

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	66685	66.685
Thin QR / faer 96 × 32 · f64	65530.00000000001	65.53000000000001
Thin QR / faer 256 × 64 · f64	447474	447.474
Thin QR / Kernix 256 × 64 · f64	693830	693.83
Thin QR / NumPy 256 × 64 · f64	565441	565.441
Thin QR / nalgebra 256 × 64 · f64	585304	585.304
Cholesky factorization / faer 64 × 64 · f64	13256	13.256
Cholesky factorization / Kernix 64 × 64 · f64	27374	27.374
Cholesky factorization / nalgebra 64 × 64 · f64	20391	20.391
Cholesky factorization / NumPy 64 × 64 · f64	17671	17.671

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	277143	277.143
Cholesky factorization / faer 128 × 128 · f64	65268	65.268
Cholesky factorization / NumPy 128 × 128 · f64	85267	85.267
Cholesky factorization / nalgebra 128 × 128 · f64	128952	128.952
Cholesky factorization / nalgebra 256 × 256 · f64	1031412	1031.412
Cholesky factorization / faer 256 × 256 · f64	404576	404.576
Cholesky factorization / NumPy 256 × 256 · f64	745574	745.574
Cholesky factorization / Kernix 256 × 256 · f64	769282	769.282
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4983	4.983
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8368	8.368

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3536	3.536
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7255	7.255
Model inference / Candle batched normalized/sliced MLP · f64	5335	5.335
Model inference / Kernix batched normalized/sliced MLP · f64	3031	3.031
Model recording + forward / Candle batched normalized/sliced MLP · f64	7600	7.6
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3856	3.856
Model backward / Kernix prepared graph → fresh gradients · f64	4200	4.2
Model backward / Candle prepared graph → fresh gradients · f64	21198	21.198
Parameter update / Kernix transactional SGD, fixed gradients · f64	728	0.728
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16415	16.415

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46718	46.718
Row softmax / ndarray 256 × 64 · f64	97274	97.274
Row softmax / NumPy 256 × 64 · f64	134696	134.696
Row softmax / Candle 256 × 64 · f64	110506	110.506
Row softmax / Kernix 256 × 64 · f64	113009	113.009
Create zeros / faer 64 × 64 · f32	320	0.32
Create zeros / Candle 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / Kernix 64 × 64 · f32	208	0.208
Create zeros / NumPy 64 × 64 · f32	528	0.528

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / Candle 64 × 64 · f32	336	0.336
Create ones / NumPy 64 × 64 · f32	1528	1.528
Create ones / Kernix 64 × 64 · f32	256	0.256